

Modelling of Large-Scale Circulation in Atmosphere and Ocean
(MO8004)

Rossby Waves

Lab 3

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Author: Fredrik Bergelv (fredrik.bergelv@live.se)

Supervisor: Verena Jung

Questions

Theory

1. State the necessary assumptions that are needed to study Rossby waves.
2. Write down the dispersion relation of Rossby waves, the equation for the group velocity, and the phase speed. Discuss the dispersion of Rossby waves.
3. Discuss in your own words the term dispersion.
4. From the equations in Question 2, discuss the role of the wave number k in relation to the Rossby radius R . Which cases can be distinguished?
5. State the equation for $\partial h / \partial t$ with variable depth

$$D = D_0 + \alpha y$$

and indicate the two additional terms that need to be implemented.

6. State the relation where α cancels the β -effect.
7. Why do we have to assume reduced gravity in this laboratory?

Experiments

Set up the model as described in the laboratory description for Laboratory 3B and follow the instructions for the experiments.

2.1 The role of wave number

1. List six values of kR and the corresponding L_w that sample the characteristic shape of the zonal Rossby wave phase speed c_x and group speed c_{gx} relations, assuming

$$L_w = \frac{1.3\sqrt{g'H}}{kR\sqrt{5}f_0}.$$

Explain your choice.

2. Plot the theoretical zonal phase speed c_x and group speed c_{gx} as a function of kR , and highlight the values corresponding to your choices in Question 1.
3. Choose three kR values representing the characteristic Rossby wave regimes and provide Hovmöller diagrams at the center of the y -axis. Estimate the numerical zonal phase and group velocities and plot them as lines in the diagrams. Add the theoretical zonal phase and group speeds as lines in the Hovmöller diagrams.
4. Add the numerical estimates of your three characteristic waves into the plot from Question 2. (It is acceptable to combine Questions 2 and 4 into one final figure.)

2.2 The role of topography

1. Fix β and choose a value of kR for experiments with varying α . Plot Hovmöller diagrams of at least three representative waves on the α -plane. Add the theoretical zonal phase and group speeds to the plots and explain your parameter choices.
2. Plot the waves shown in Question 1 as xy cross-sections at an appropriate time.
3. For two of your α -value experiments (excluding the cancelling case), plot the theoretical c_g and c_x as a function of kR and include the numerical estimates from the experiments. Compare the results with the plot from Question 2.1.4.

1 Theory

The dispersion relation for a Rossby wave with meridional varying topography is

$$\omega = -k \frac{\beta - \frac{f_0}{H_0} \alpha}{k^2 + l^2 + \frac{1}{R^2}} \quad (1)$$

v) The normal convergence-equation in the shallow water equations is

$$\frac{\partial \eta}{\partial t} + \frac{\partial(uh)}{\partial x} + \frac{\partial(vh)}{\partial y} = 0. \quad (2)$$

Normally we use that $h = H_0 + \eta$, where $\eta \ll H_0$. But now we instead use a varying topography with $h = H_0 + \alpha y + \eta$, where $\eta \ll H_0$ and $\alpha y \ll H_0$. Using this on Equation 2 we obtain

$$\Rightarrow \frac{\partial \eta}{\partial t} + H_0 \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + \underbrace{\alpha y \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right)}_{\text{new topographic terms}} + \alpha v = 0. \quad (3)$$

2 Experiments

2.1 The role of wave number

i) We know that the width of the initial gaussian, L_w is related to the wave number and Rossby radius as

$$L_w = 1.3 \frac{\pi}{kR} \frac{1}{\sqrt{5}} \frac{\sqrt{g'H}}{f_0}.$$

Furthermore, from the theoretical solution we know the dispersion relation as Equation 1, where we observe that without the topography and a wave solution in the x -direction, gives $\omega = 0$ at $kR = \pm 1$. From this we can sample 6 points as

kR	L_w (km)	% of domain
0.50	723.61	10.3372
1.00	361.80	5.1686
1.73	209.13	2.9876
3.00	120.60	1.7229
5.00	72.36	1.0337
7.00	51.69	0.7384

Table 1: This table displays the kR -points which were selected for analysis, their corresponding L_w and that L_w expressed as a presented of the domain length (7×10^6 m).

The points in Table 1 were selected to sample as many points from the theoretical shape (which can be seen in Figure 2). Since the dispersion relating is symmetric in the plus/minus domain, only positive points were chosen (since negative points would just give the antisymmetric speeds). The especially important points [0.5, 1, 1.73, 3, 5, 7], which were argued from Equation 1. The point $kR = 0.5$ was chosen since this corresponds to the long-wave limit, where we expect to see less dispersive waves which move more slowly. The point $k = \frac{1}{R}$ (i.e. $kR = 1$) was chosen since at this point we expect to see a stationary wave packet. The point $kR = 1.73$ was chosen since this is the point where we would observe the strongest eastward group velocity. The points [3, 5, 7] were chosen to examine the short-wave limit where we expect to see a convergence towards stationary waves.

iii) A Hovmöller diagram showcasing the wave for the different kR -point from Table 1 can be seen in Figure 1. There one can see the group-speed as the direction between the peaks and valleys. The phase

speed can be seen as the temporal line of peaks from the initial position. One can observe good agreement from the values where $kR \geq 1$. Especially good result can be observed when $kR = 1$, where the theoretical and numerical phase/group-speeds really align. The point $kR = 0.5$ clearly gave the worst result, where a large difference between the numerical and theoretical speeds can be observed.

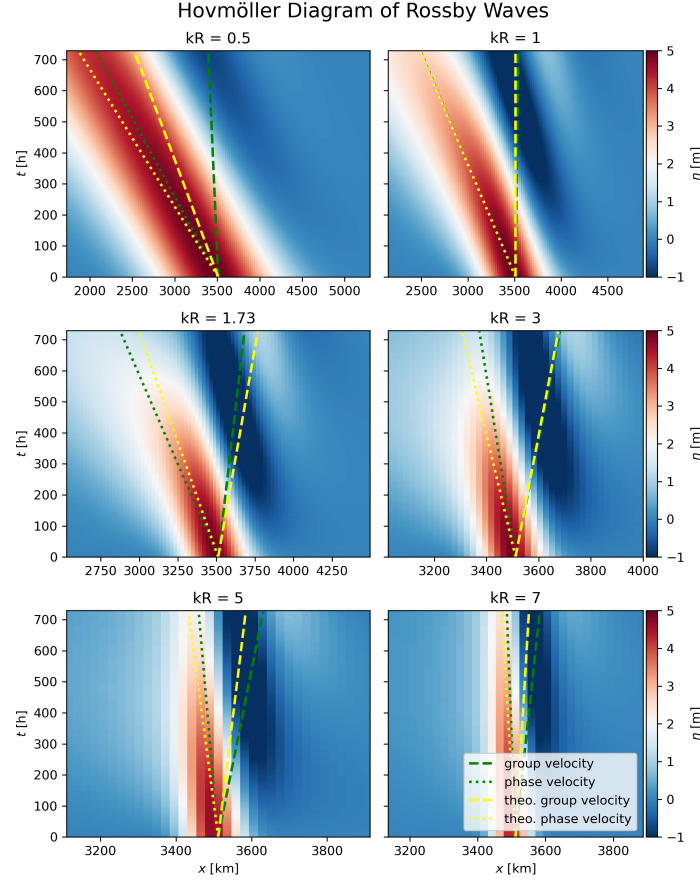


Figure 1: This figure showcases the phase and group propagation for the different kR -point from Table 1.

ii, iv) One can observe the differences between the numerical and theoretical phase/group speeds together with the theoretical curves in Figure 2. There one can observe more clearly the differences seen in Figure 1. Again one can observe good agreement when $kR \geq 1$ and bad agreement when $kR = 0.5$. The reason for this long-wave mismatch can be described by the nature of long numerical Rossby waves.

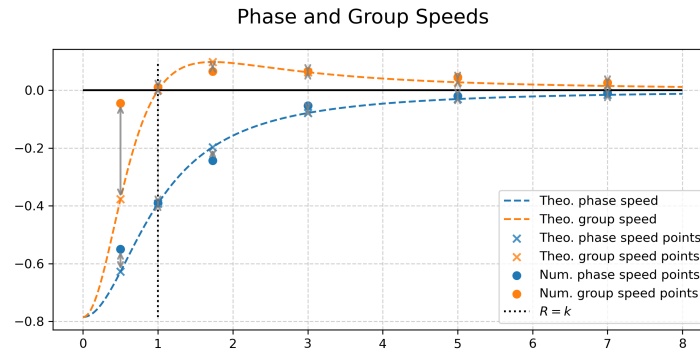


Figure 2: This figure shows the theoretical phase and group speeds together with the theoretical and numerical speeds for the kR -point from Table 1.

2.2 The role of topography

- i) Choose a kR where we see stuff, i.e. do not choose extremes.
- ii) Do a normal contour-plot. for the different α .
- iii)

A The Source Code

The source code for this lab can be found on GitHub:

`https://github.com/FredrikBergelv/
Modelling-of-Large-Scale-Circulation-in-Atmosphere-and-Ocean`